MANISH KUMAR BHATT

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A highly accomplished Systems Architecture and Hardware Engineering Leader with over 20 years of experience in developing scalable platforms and driving product innovation across commercial and industrial electronics. Brings deep expertise in end-to-end product development, backed by a strong focus on modern engineering practices, system-level thinking, and technology adoption to enable business growth. Proven track record of leading R&D initiatives and contributing to high-impact engineering teams within globally recognized organizations.

Professional Summary

Strategic hardware leader with expertise in driving R&D, product platforming, and scalable architectures aligned to long-term business objectives. Proven ability to define and govern end-to-end hardware architecture, system design, and execution across multiple programs with predictable delivery. Strong advocate of systems-driven development with full traceability from requirements to validation, ensuring compliance and quality. Experienced in leading enterprise technical governance reviews and managing technical scope, risks, and global standards adherence. Skilled in portfolio planning, including schedules, resource capacity, and budget ownership. Adept at cross-functional leadership, aligning engineering, supply chain, manufacturing, and regulatory teams for scalable and cost-effective product delivery. Deep technical expertise in high-speed embedded platforms and complex systems using SoCs, FPGAs, and advanced interfaces. Passionate about building high-performance teams, fostering innovation, and driving engineering excellence. Hands-on in critical phases such as board bring-up, design reviews, and complex debugging. Committed to delivering robust, compliant products through strong validation strategies, EMI/EMC readiness, and KPI-driven execution.

Career Timeline

- May 2022 - Present — **Lead Hardware Engineer**, Allegion Technology PVT LTD
- Feb 2021 - Apr 2022 — **Hardware Architect**, Pixcellence Technology PVT LTD
- Jul 2020 - Jan 2021 — **Sr. Manager (Hardware)**, VVDN Technology
- Mar 2007 - Feb 2020 — **Sr. Lead Engineer (R&D Hardware)**, Barco Electronic System
- Dec 2003 - Mar 2007 — **Scientist B**, CSIO (CSIR LAB) Chandigarh

Work Experience

May 2022 - Present

Lead Hardware Engineer

Allegion Technology PVT LTD (Bangalore)

- Spearheaded end-to-end hardware development for New Product Development (NPD) and asset development programs, ensuring timely execution and adherence to project milestones.
- Defined technical scope and ensured compliance, maintaining traceability to industry standards, regulations, and measurable verification metrics.
- Drove requirements-based product development through customer need analysis, systems thinking, and structured SRD to FRD decomposition, aligning product features with customer expectations.

Soft Skills

Leadership · Mentorship · Communication · Problem-solving

Technical Skills

- End to end PDLC NPD/NPI
- Requirements Based Product Development
- High Speed Board Design
- High Speed Interfaces (USB, MIPI, DVI, HDMI, Display port)
- Embedded Bus Protocol (I2C, SPI, USB JTAG)
- Wireless interfaces (NFC LoRA WIFI Bluetooth)
- IDDE (TI code composer studio, Visual DSP)
- Requirements Management Tool (Devops, JIRA)
- PLM/ERP Tools (Windchill, BAAN, SAP)
- Simulation (SI PI Hyperlynx)

Core Competencies

- Requirements Based Product Development
- interface control document (ICD)
- Requirements Life cycle Management
- Embedded System Design
- Product Platforming/Reuse strategy
- Program & Portfolio Management (NPD/NPI)
- Validation Strategy (DVP&R, DFMEA, IV&V)
- System Architecture & Requirements Management
- Stakeholder & Executive Communication
- Cross-Functional Leadership (HW, SW, QA, Supply Chain)
- Program Planning, Scheduling & Execution

Certifications

- Managed requirements lifecycle using Azure DevOps, facilitating traceability, change control, and cross-functional alignment, resulting in streamlined development processes.
- Led risk identification, assessment, and mitigation planning, ensuring stakeholder alignment and predictable execution across hardware projects.
- Facilitated global collaboration with supply chain, sourcing, and regulatory bodies to support compliant and scalable hardware development, mitigating potential roadblocks and ensuring adherence to industry standards.
- Developed and executed Design Validation Plan & Report (DVP&R) and Design Failure Mode and Effects Analysis (DFMEA) plans to validate design robustness, reliability, and quality, reducing potential failure points and enhancing product performance.
- Defined asset strategy and product platforming approaches to enable reuse, scalability, and portfolio optimization, enhancing efficiency and reducing development costs.
- Prepared and maintained Interface Control Documents (ICD) for asset and system-level integration, ensuring seamless interaction between components.
- Partnered with cross-functional architects (systems, firmware, mechanical, compliance) to advance the technical excellence of the Embedded Systems group, fostering innovation and enhancing team capabilities.
- Resolved both intra-functional and inter-functional technical conflicts, aligning teams toward unified project and business objectives.
- Oversaw Independent Verification & Validation (IV&V) and system verification activities across the V-model, ensuring independent and objective assessment of requirements compliance.
- Led and mentored a cross-functional team of 5 engineers, fostering technical delivery, skill development, and execution excellence.

Achievements:

- Successfully delivered NPD and asset development programs, achieving milestone reviews with gate exits (ADR, SYSDR, HWDR) within established timelines and budgets.
- Enhanced design robustness, reliability, and quality by executing DVP&R and DFMEA plans and mitigating potential failures.
- Improved project execution and team performance through effective leadership and mentorship, resulting in improved technical delivery and skill development for team members.

Feb 2021 - Apr 2022

Hardware Architect

Pixcellence Technology PVT LTD

- Orchestrated brainstorming sessions and architected hardware solutions aligning with the product requirements for Smart Video IoT Doorbell and Video Outdoor Camera developments.
- Directed and managed the complete product development lifecycle of video doorbell products, ensuring adherence to project timelines, technical specifications, and budgetary constraints.
- Applied expertise in Sigma Star chipset SOC530/SOC837, leveraging its capabilities for high-end embedded products in the IoT domain.
- Integrated wireless protocols LoRa, NFC, and WiFi, optimizing connectivity and communication performance for developed products.
- Addressed and resolved EMI/EMC issues during schematic design, layout, and stack-up phases, ensuring compliance with regulatory standards like FCC.
- Managed prototyping, BOM preparation, test plan preparation, and execution, to facilitate delivery to mass production with minimal defects.
- Administered project milestone control, risk management, and issue tracking, guaranteeing efficient project execution and timely resolution of any encountered challenges.
- Fostered effective project planning and coordination with customers and internal teams, maintaining transparent communication and alignment throughout the project lifecycle.

Achievements:

- Successfully navigated product from concept, design and development to prototyping and mass production.
- Resolved EMI/EMC compliance issues through design, layout and stack-up optimization, securing FCC certifications.

- **Working with Dxdesigner"** (by Mentor Graphics)
- **HyperLynx Signal Integrity Training** (by Mentor Graphics")
- **EMC For Electronic Engineer "** (By Keith Armstrong)
- **Techniques for Grounding Shielding & Transmission Lines** (PCB West - By Denial Beeker)
- **Requirements Based Product Development** (By Dr Dan Surber)

Education

- **M.TECH. in Electronics and Communication**
  National institute of technology Kurukshetra
  2002 - 2005
- **B.Tech. in Electronics and Communication**
  Agra University
  1997 - 2001

Languages

English Hindi

Hobbies

Technology Innovation Gadgets DIY Electronics

Achievements

- Selected in DRDO Scientist Entry Test (2003)
- Selected CSIR Scientist Entry Test (2003)
- Achieved 97th Percentile in GATE (2002)
- Achieved 96.06th Percentile in GATE (2003)

- Enhanced product functionality and reliability by integrating LoRa, NFC, and WiFi to deliver a competitive edge.

Jul 2020 - Jan 2021

Sr. Manager (Hardware)

VVDN Technology

- Managed product development projects from initial design through testing, providing strategic direction and oversight.
- Oversaw all phases of the Hardware Development Life Cycle, including requirements analysis, component selection, design, development, testing, and documentation.
- Orchestrated product development in coordination with PCB and mechanical CAD, signal integrity, thermal and power design, safety, EMC, DVT, and PVT teams.
- Facilitated Electronics Design, Analysis, PCB layout & routing, manufacturing & assembly, testing, verification & Integration support ensuring consistent end-to-end workflow.
- Supported proto builds, pilot builds, and mass production activities, facilitating smooth transitions between development phases.
- Leveraged expertise in Qualcomm chipset QCS605 PIMIC670, PIMIC 670L etc. for high-end embedded products, ensuring optimal performance and features.
- Applied in-depth knowledge of high-speed protocols (USB, DP, HDMI, MIPI-CSI, MIPI-DSI) and low-speed protocols (SPI, I2C, UART, I2S).

Achievements:

- Successfully managed and executed hardware development projects from concept to mass production.
- Ensured consistent, high-quality product development by coordinating with PCB, mechanical CAD and other engineering teams.
- Integrated high-speed and low-speed protocols with expertise thus improving product compatibility and functionality.

Mar 2007 - Feb 2020

Sr. Lead Engineer (R&D Hardware)

Barco Electronic System

- Spearheaded brainstorming sessions to define hardware architecture in alignment with top-level project requirements.
- Conducted hands-on hardware system emulation, integration, prototype bring-up, and debugging to optimize system functionality and performance.
- Engineered and developed high-speed boards for video acquisition (DVI, HDMI, DP), enhancing video processing capabilities.
- Defined Firmware (FMW) architecture and developed FMW for MSP/TIVA based Microcontrollers for Video Wall Energy Star Compliance.
- Performed design validation and resolved issues ensuring successful system integration and stability.
- Interacted with firmware, optics, mechanical, and FPGA team leads to define block-level design, fostering team collaboration and design consistency.
- Designed processor-based systems and interfaced peripherals (DDR3, FPGA) for video processing of DLP-based projection systems.
- Conducted reviews and provided suggestions on component selection and circuit design for DLP/ PANEL based high-processing embedded systems, optimizing system performance.
- Prepared layout guidelines and collaborated with layout teams to ensure accurate and effective board designs.
- Led in-depth design reviews to ensure quality and adherence to design specifications.
- Managed activities like component selection, circuit design, and signal integrity analysis to maintain signal quality and system stability.
- Consulted with various Agency Approval Standards (UL, CE, CB) ensuring product compliance.
- Orchestrated value engineering exercises for cost reduction of products and increased product value, enhancing product competitiveness.
- Developed detailed project documentation maximizing team effectiveness and cross-functional understanding.

Achievements:

- Implemented hardware solutions enhancing video processing capabilities for high-speed video boards (DVI, HDMI, DP)
- Improved energy efficiency by defining and developing Firmware for MSP/TIVA based Microcontrollers for video-wall energy star compliance.
- Reduced product costs and increasing product value by leading value engineered exercises.

Dec 2003 - Mar 2007

Scientist B

CSIO (CSIR LAB) Chandigarh (Chandigarh)

Microcontroller (8bit/16bit) based embedded computing system design with peripheral interfacing for cockpit display system for HJT 36 Trainer Aircraft and Light Combat Aircraft. ADSP2181 based embedded Hardware design for character and symbol generation on CRT for Head up Display (HUD) forHJT 36 Trainer Aircraft. Firmware development with C and assembly language programming with Keil and Visual DSP for cockpit display system. Involved in various certification agencies for Airworthiness.

- Engineered microcontroller (8bit/16bit) based embedded computing system designs with peripheral interfacing for cockpit display systems in the HJT 36 Trainer Aircraft and Light Combat Aircraft.
- Developed ADSP2181 based embedded hardware designs for character and symbol generation on CRT for Head-up Displays (HUD) in the HJT 36 Trainer Aircraft.
- Developed firmware using C and assembly language programming with Keil and Visual DSP for cockpit display systems, thus optimizing system performance.
- Participated in various certification processes with airworthiness agencies to ensure compliance and safety of designed systems, and adherence to specifications related to airworthiness.

Achievements:

- Contributed to airworthiness certification of cockpit display systems highlighting adherence to industry safety standards.
- Improved cockpit display capability by successfully developed firmware using C with assembly.